# 【10.8| Economist】One way or another 无论怎样

**难度评级：★★★★☆**

**【考研那些事儿之科技系列】**

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# Alternatives to GPS are around the corner

# GPS 的替代方案即将来临

【 学习流程：第一遍请先看原文自己精读，第二遍对照着本精读文档进行理解。文章标注的单词为重点单词，只提供相关注释，想要详细释义麻烦同学查一下词典哦！另外文章底色为灰色的句子为今天的翻译作业，今日文档中的句子分析模块会提供答案，同学们请先思考再看句子分析～】

【Para.1】

The first **casualty** of war is truth. But a reliable GPS signal comes a close second. **Jamming** signals from a global navigation satellite system (GNSS), such as America's GPS, is so easy it has become common when the shooting starts. Stronger signals from newer kit can make jamming harder, but the potential scale of the threat now looks much greater than previously thought. A paper posted in June by a team led by Todd Humphreys of the University of Texas, Austin, traced scores of broad GPS disruptions across Europe and elsewhere since 2019 to a **constellation** of Russian satellites. Jamming or even **inadvertent** interference from space, the research suggests, could affect an entire continent.

- **around the corner phrase.** 即将来临：指某事很快就会发生。
- **casualty n.**/ˈkæʒuəlti/ 1) a person who is killed or injured in war or in an accident（战争或事故的）伤员，亡者，遇难者

  【考研大纲词汇】

  road casualties 交通事故伤亡人员

2) a person that suffers or a thing that is destroyed when sth else takes place 受害者；毁坏物；损坏物

SYN victim

Small shops have been a casualty of the recession. 小商店在经济萧条中深受其害。

3) the part of a hospital where people who need urgent treatment are taken 急诊室

The victims were rushed to casualty. 受伤者被迅速送往急救室。

- **jamming n.**/ˈdʒæmɪŋ/ 干扰，电子干扰（电子、通信等领域，亦译“抑制”）；堵塞，卡住（交通、机械、纸张等）
- **constellation n.**/ˌkɒnstəˈleɪʃn/ 1) a group of stars that forms a shape in the sky and has a name 星座

2) a group of related ideas, things or people 一系列（相关的想法、事物）；一群（相关的人）

a constellation of Hollywood talent 一群好莱坞天才

- **inadvertent adj.**/ˌɪnədˈvɜːt(ə)nt/ 无意的，非故意的，无意中做的

战争的第一牺牲品是真相。但稳定的 GPS 信号紧随其后。

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干扰全球导航卫星系统（GNSS，如美国的 GPS）十分容易，以至于战火一旦燃起，这便成为常态。新型装备发出的更强型号虽然能增加干扰难度，但这一威胁的潜在规模似乎远超此前的想象。由德克萨斯大学奥斯汀分校的托德·汉弗莱斯领导的一个团队在 6 月发表的一篇论文，将自 2019 年以来欧洲和其他地区数十次大范围 GPS 中断追溯到一组俄罗斯卫星。该研究表明，来自太空的干扰甚至无意的干涉，可能影响整个大陆。

【Para.2】

One alternative to GNSS is simply to mount cameras on a craft and use software to keep track of the **terrain** below. But that works only over land, and can be **stymied** by cloud cover, **haze** or darkness. So two other, quite different, methods are under investigation. One employs **quantum** phenomena—the **angular** momentum of **photons** and the **spin** of electrons—to follow **magnetic** maps of Earth's surface. The other establishes an observer's position by tapping **ambient** radio waves broadcast for other purposes. Neither method can yet match the precision of GNSS, but both are becoming credible backups.

- **terrain n.**/təˈreɪn/ used to refer to an area of land when you are mentioning its natural features, for example, if it is rough, flat, etc.地形；地势；地带

  difficult/rough/mountainous, etc. terrain 难以通过的地带、崎岖不平的地形、山地等

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- **stymie v.**/ˈstaɪmi/ to prevent sb from doing sth that they have planned or want to do; to prevent sth from happening 阻挠；阻碍；阻止；妨碍

  SYN foil

- **haze n.**/heɪz/ 1) air that is difficult to see through because it contains very small drops of water, especially caused by hot weather（尤指热天引起的）薄雾，霾

  a heat haze 热天的雾气

  2) air containing sth that makes it difficult to see through it（烟尘等的）雾霭，烟雾

  a haze of smoke/dust/steam 烟/尘/蒸汽雾

  3) a mental state in which your thoughts, feelings, etc. are not clear 迷蒙；迷糊

  an alcoholic haze 喝醉酒的迷糊

- **quantum n.**/ˈkwɒntəm/ a very small quantity of electromagnetic energy 量子

- **angular adj.**/ˈæŋgjələ(r)/ 1) thin and without much flesh so that the bones are noticeable 瘦骨嶙峋的；骨瘦如柴的

  an angular face 瘦削的脸

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2) having angles or sharp corners 有棱角的；有尖角的

a design of large angular shapes 大棱角形的图案

- **photon n.**/ˈfəʊtɒn/ a unit of electromagnetic energy 光子；光量子
- **spin n.**/spɪn/ 1) a very fast turning movement 高速旋转【考研大纲词汇】

  the earth's spin 地球的自转

  2) if an aircraft goes into a spin , it falls and turns round rapidly（飞机的）尾旋，螺旋式下坠

  3) a short ride in a car for pleasure 兜风

  Let's go for a spin. 咱们兜风去。

  4) a way of presenting information or a situation in a particular way, especially one that makes you or your ideas seem good（尤指有利于自己的）导向性陈述

  Politicians put their own spin on the economic situation. 政治家们对经济形势各执一词。
- **magnetic adj.**/mægˈnetɪk/ 1) behaving like a magnet 像磁铁的；有磁性的【考研大纲词汇】

  magnetic materials 磁性材料

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2) connected with or produced by magnetism 磁的；磁性的

magnetic properties/forces 磁性/力

3) that people find very powerful and attractive 富有吸引力的；有魅力的

a magnetic personality 富有魅力的个性

- **ambient adj.**/ˈæmbiənt/ 1) relating to the surrounding area; on all sides 周围环境的；周围的

ambient temperature/light/conditions 周围的温度/光线/环境

2) creating a relaxed atmosphere 产生轻松氛围的

soft, ambient lighting 轻松柔和的照明

GNSS 的一种替代方案是简单地在飞行器上安装摄像头，并使用软件跟踪下方的地形。但这只在陆地上空有效，并且可能被云层、雾霾或黑暗所阻碍。因此，另外两种截然不同的方法正在研究中。一种利用量子现象——光子的角动量和电子的自旋——来跟踪地球表面的磁场分布。另一种则是借助其他用途的环境无线电波，来确定观测者的位置。这两种方法目前都还不能与 GNSS 的精度相媲美，但两者都在成为可靠的备用方案。

【Para.3】

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Many rocks are weakly magnetic, and their distribution in Earth's crust results in irregularities in the planet's magnetic field. The resulting landscape of **anomalies** is detectable under almost any conditions, and equally well over land and sea. Fit a plane with a magnetometer capable of spotting these anomalies, and a magnetic navigation (MagNav) system that can match them to existing maps, and its position can be accurately **plotted**. Last year Q-CTRL, an Australian firm, demonstrated a MagNav system that was more accurate than a leading GPS alternative in a trial flight over New South Wales.

- **anomaly n.**/əˈnɒməli/ a thing, situation, etc. that is different from what is normal or expected 异常事物；反常现象

  the many anomalies in the tax system 税制中的许多破格现象

- **plot v.**/plɒt/ 1) to make a secret plan to harm sb, especially a government or its leader 密谋；暗中策划【考研大纲词汇】

  SYN conspire

  They were accused of plotting against the state.他们被指控阴谋叛乱。

  2) to mark sth on a map, for example the position or course of sth（在地图上）画出，标出

  The earthquake centres had been plotted on a world map. 地震震中均被标示在一张世界地图上。

  3) to make a diagram or chart from some information 绘制

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（图表）

We carefully plotted each patient's response to the drug on a chart. 我们仔细绘出了每个病人对这种药物反应的图表。

许多岩石是具有弱磁性的，它们在地壳中的分布导致地球磁场的不规则性。由此产生的磁异常分布几乎在任何条件下都可检测到，在陆地和海洋上同样如此。在一架飞机上安装能够发现这些异常的磁力仪，以及一套能够将它们与现有地图匹配的磁导航（MagNav）系统，其位置就可以被精确标绘。去年，澳大利亚公司 Q-CTRL 在新南威尔士州的一次试飞中展示了一套 MagNav 系统，其精度超过了一个领先的 GPS 替代方案。

【Para.4】

Q-CTRL's sensor measures the field's local strength using a device called an optically pumped magnetometer (OPM). This works by firing a laser into a cloud of **rubidium** atoms. If the **laser** beam is tuned correctly, the atoms' electrons enter a state in which they are extremely sensitive to external magnetic fields. Another laser can then be used to track the behaviour of the electrons and work out the ambient magnetic field, allowing the system's software to work out where on the map it is.

- **rubidium n.**/rʊˈbɪdiəm/ a chemical element. Rubidium is a rare soft silver-coloured metal that reacts strongly with water and burns when it is brought into contact with air. 金属铷

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- **laser n.**/ˈleɪzə(r)/ a device that makes a very strong line of controlled light (= with rays that are parallel and of the same wavelength) 激光器【考研大纲词汇】

  a laser beam 激光束

Q-CTRL 的传感器借助一种称为光泵磁力仪 (OPM) 的设备测量磁场的局部强度。其工作原理是向一团铷原子发射激光。如果激光束调谐得当，原子中的电子会进入一种对外部磁场极其敏感的状态。然后可以用另一束激光跟踪电子的行为并测算出环境磁场，从而使系统的软件能够计算出它在地图上的位置。

【Para.5】

Other firms are betting on a different type of sensor capable of showing not only the field's strength, as Q-CTRL's sensors already do, but also its direction. They believe these sensors, if used to **navigate** against the **vector** maps they help produce, could allow an observer to more accurately **pinpoint** their position.

- **navigate v.**/ˈnævɪgeɪt/ 1) to find your position or the position of your ship, plane, car etc. and the direction you need to go in, for example by using a map 导航；确定（船、飞机、汽车等）的位置和方向【考研大纲词汇】

  to navigate by the stars 根据星辰确定航向

2) to find the right way to deal with a difficult or complicated situation 找到正确方法（对付困难复杂的情况）

We next had to navigate a complex network of committees. 我们下一步必须设法使各级委员会予以通过。

- **vector n.**/ˈvektə(r)/ 1) a quantity that has both size and direction 矢量；向量

  2) an insect, etc. that carries a particular disease from one living thing to another（传染疾病的）介体，载体

  3) a course taken by an aircraft（航空器的）航线

- **pinpoint v.**/ˈpɪnpɔɪnt/ to find and show the exact position of sb/sth or the exact time that sth happened 明确指出，确定（位置或时间）

  He was able to pinpoint on the map the site of the medieval village. 他能在地图上准确找出那个中世纪村庄的位置。

其他公司则押注于一种不同类型的传感器，它不仅能像 Q-CTRL 的传感器那样显示磁场强度，还能显示其方向。他们相信，如果利用这些传感器和他们帮助生成的矢量地图进行导航，就可以让观察者更精确地确定自己的位置。

【Para.6】

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One such firm is SBQuantum, a Canadian **outfit**. Its device is called an NV-diamond magnetometer which, as the name suggests, has a diamond at its heart. Diamonds are not normally magnetic, but can be made so by creating what are known as **nitrogen-vacancy** (NV) centres in their **crystal lattices**. Once the diamond is placed in a magnetic field, its lattice provides a 3D **grid** against which the electrons' behaviour can be measured. Accurate readings of field strength and direction can then be taken.

- **outfit n.**/ˈaʊtfɪt/ 1) a group of people working together as an organization, business, team, etc.团队；小组；分队【考研大纲词汇】

  a market research outfit 市场调查组

  2) a set of clothes that you wear together, especially for a particular occasion or purpose 全套服装，装束（尤指为某场合或目的）

  a wedding outfit 一套结婚礼服

- **nitrogen n.**/ˈnaɪtrədʒən/ a chemical element. Nitrogen is a gas that is found in large quantities in the earth's atmosphere.氮；氮气【考研大纲词汇】

- **vacancy n.**/ˈveɪkənsi/ 1) a job that is available for sb to do （职位的）空缺；空职；空额

  job vacancies 职位空缺

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2) a room that is available in a hotel, etc.（旅馆等的）空房，空间

I'm sorry, we have no vacancies. 对不起，我们这里客满。

3) lack of interest or ideas 无兴趣；无主意；空虚

SYN emptiness

- **crystal n.**/ˈkrɪstl/ 1) a small piece of a substance with many even sides, that is formed naturally when the substance becomes solid 结晶；晶体【考研大纲词汇】

  ice/salt crystals 冰/盐的结晶体

  2) a clear mineral, such as quartz , used in making jewellery and decorative objects 水晶（如石英，用于制作珠宝饰物）

  a pair of crystal earrings 一对水晶耳环

- **lattice n.**/ˈlætɪs/ a structure that is made of strips of wood or metal that cross over each other with spaces shaped like a diamond between them, used, for example, as a fence; any structure or pattern like this 格子木架，格子金属架，格栅（用作篱笆等）；斜条结构；斜格图案

- **grid n.**/grɪd/ 1) a pattern of straight lines, usually crossing each other to form squares 网格；方格

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2) a system of electric wires or pipes carrying gas, for sending power over a large area（输电线路、天然气管道等的）系统网络；输电网；煤气输送网

其中一家是加拿大公司是 SBQuantum。它的设备被称为 NV 金刚石磁力仪，顾名思义，其核心是一颗金刚石。金刚石通常没有磁性，但可以通过在其晶格中创建所谓的氮空位（NV）中心来使其具有磁性。一旦金刚石被置于磁场中，其晶格就提供了一个 3D 网格，可以据此测量电子的行为。然后就可以准确读取磁场强度和方向。

【Para.7】

For now, the main drawback of both these approaches is a **dearth** of good maps of the **crustal** field. Those that are available tend to be **mutually** inconsistent and often lack the accuracy needed for navigation. Some firms, SBQuantum among them, therefore fly survey missions to create their own maps, while others, such as SandboxAQ, a Californian company, propose improving those that already exist.

- **dearth n.**/dɜːθ/ a lack of sth; the fact of there not being enough of sth 缺乏；不足

  SYN scarcity

  There was a dearth of reliable information on the subject.

  关于这个课题缺乏可靠资料。

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- **crustal adj.**/ˈkrʌst(ə)l/ 地壳的；（行星、卫星等）外壳的；壳的
- **mutually adv.**/ˈmju: tʃuəli/ felt or done equally by two or more people 相互地；彼此；共同地

  a mutually beneficial/supportive relationship 互惠/互助的关系

目前，这两种方法的主要缺点都是缺乏优质的地壳磁场地图。现有的地图往往相互矛盾，并且常常满足导航所需的精度。因此，一些公司——包括 SBQuantum——开展飞行勘测任务来创建自己的地图，而其他公司，如加利福尼亚公司 SandboxAQ，则提议改进已有的地图。

【Para.8】

Meanwhile, others are developing ways to derive position from existing satellite radio signals. America's Federal Communications Commission has pressed for more work on such "signals of opportunity", and a few teams are giving it a shot. One technique is to measure Doppler shifts. As a satellite approaches and then **recedes**, the frequency of its transmissions appears to rise and then drop, much like the **whistle** of a passing train. The closer an observer is to a satellite's path, the more rapid the shift. Information about satellite orbits can then be combined with frequency measurements to estimate position.

- **recede v.**/rɪˈsi: d/ 1) to move gradually away from sb or away from a previous position 逐渐远离；渐渐远去【考研大纲

词汇】

The sound of the truck receded into the distance. 卡车的声音渐渐在远处消失了。

2) to become gradually weaker or smaller 逐渐减弱；慢慢变小

The pain was receding slightly. 疼痛正在一点一点地减弱。

3) to stop growing at the front of the head（头顶前部）头发停止生长，变秃

a middle-aged man with receding hair/a receding hairline 发际后移的中年男子

- **whistle n.**/ˈwɪsl/ the sound made by blowing a whistle 哨子声【考研大纲词汇】

与此同时，其他人正在开发从现有卫星无线电信号中推导位置的方法。美国联邦通信委员会已敦促在这方面对“机会信号”进行更多研究，一些团队正在尝试。一种技术是测量多普勒频移。当一颗卫星靠近然后远离时，其传输频率似乎先上升然后下降，很像一列经过的火车的汽笛声。观察者越接近卫星的路径，频移就越快。随后可以将有关卫星轨道的信息与频率测量相结合来估计位置。

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【Para.9】

One pioneer is ASPIN Lab, a research outfit at Ohio State University. In a demo on a **cruise** ship off western Greenland on August 24th 2024, its system, called Matrix, processed transmissions from 21 Starlink and OneWeb broadband satellites passing overhead. Given a starting position and speed, Matrix placed the ship just 27 metres from its true position.

- **cruise n.**/kru:  z/ a journey by sea, visiting different places, especially as a holiday/vacation 乘船游览；航行【考研大纲词汇】

  a luxury cruise ship 豪华游轮

其中一个先行者是俄亥俄州立大学的研究机构 ASPIN 实验室。在 2024 年 8 月 24 日格陵兰岛西部附近一艘游轮上的演示中，其名为 Matrix 的系统处理了从头顶经过的 21 颗星链和 OneWeb 宽带卫星的信号传输。给定一个起始位置和速度，Matrix 将船只定位在距离其真实位置仅 27 米处。

【Para.10】

Precision, though, should improve. SpaceX, the owner of Starlink, now publishes predictions for the **orbits** of its more than 11,000 satellites. Unlike tracking data produced by America's Space Force, these take into account planned **manoeuvres**. Dr Humphreys says SpaceX's forecasts tend to be accurate to within just a couple metres of the satellites' true positions. This should allow receivers to better interpret the Doppler shifts they observe.

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- **orbit v.**/ˈɔːbɪt/ 1) a curved path followed by a planet or an object as it moves around another planet, star, moon, etc.（天体等运行的）轨道【考研大纲词汇】

  a space station in orbit round the moon 绕月球运行的一个航天站

  2) an area that a particular person, organization, etc. deals with or is able to influence（人、组织等的）影响范围，势力范围

  to come/fall/be within sb's orbit 进入/落入/属于某人的势力范围

- **manoeuvre n.**/məˈnuːvə(r)/ 1) a movement performed with care and skill 细致巧妙的移动；机动动作【考研大纲词汇】

  a complicated/skilful manoeuvre 复杂的/熟练的移动

  2) a clever plan, action or movement that is used to give sb an advantage 策略；手段；花招；伎俩

  diplomatic manoeuvres 外交策略

  3) military exercises involving a large number of soldiers, ships, etc.军事演习；作战演习

  The army is on manoeuvres in the desert. 军队在沙漠中进行

军事演习。

不过，精度有望提高。星链的母公司 SpaceX 现在发布其超过 11,000 颗卫星的轨道预测。与美国太空军提供的跟踪数据不同，这些预测考虑了预定机动。汉弗莱斯博士说，SpaceX 的预测往往精确到卫星真实位置的两米以内。这将使接收器更好地解释它们观察到的多普勒频移。

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## ➢ 今日翻译作业

(Para.3) Fit a plane with a magnetometer capable of spotting these anomalies, and a magnetic navigation (MagNav) system that can match them to existing maps, and its position can be accurately plotted.

（下一页提供句子分析）

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## ➢ 今日句子分析

Para.3 Fit a plane with a magnetometer capable of spotting these anomalies, and a magnetic navigation (MagNav) system that can match them to existing maps, and its position can be accurately plotted.

在一架飞机上安装能够发现这些异常的磁力仪，以及一套能够将它们与现有地图匹配的磁导航（MagNav）系统，其位置就可以被精确标绘。

祈使句

Fit a plane with a magnetometer capable of spotting these

谓语　宾语　with 引导的方式状语

anomalies, and a magnetic navigation (MagNav) system

定语从句　　并列分句

that can match them to existing maps, and its position can

引导词　　主语　谓语

be accurately plotted.

句子分析：

这是一个并列复合句，核心框架为：

祈使句（条件） + and + 结果分句

这种结构在英语中非常常见，相当于“如果……，那么……”或“只要……，就……”。祈使句部分提出条件或做法，and 后面的分句说明结果。

全句没有传统意义上的从句嵌套，但通过介词短语 + 形容词短语 + 定语从句形成了多层修饰，是典型的科技说明文长句。

翻译注意事项：

- 祈使句 + and 结构的转换 ：英文祈使句表条件，and 表结果；中文习惯用“只要……就……”或“如果……那么……”。 译为“只要给飞机装上……，就能……”或“给飞机装上……，其位置就能……”。
- Fit...with... 的搭配：第一个用形容词短语 capable of...，第二个用定语从句 that can...。中文定语通常前置，但过长时需拆分。 第一个译为“能够识别这些异常的磁力仪”；

第二个译为“能将这些异常与现有地图匹配的磁导航系统”。

- 科技文体的简洁性：英文用祈使句显得简洁有力，中文若直译“安装一架飞机……”会别扭。 需补充主语或改为“只要给飞机装上……”，使中文通顺。

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## ➢ 写作积累

1.(Para.1) The first casualty of war is truth. But a reliable GPS signal **comes a close second**.

**come a close second ：**意为“紧随其后，位居第二”。作者借用“战争第一牺牲品是真相”这一名言，指出可靠的 GPS 信号是第二个最容易在战争中受损的事物。它强调的不是“第二名”本身，而是紧随第一名之后的紧密程度。源自赛马、赛跑等竞技语境，指某选手或某事物紧跟在第一名之后到达或排名第二，差距很小。

◆ 类似搭配

- come a poor second　远远落在第二（差距大）
- rank second only to...　仅次于...
- come a distant second　远远落在第二

2.(Para.7) For now, the main drawback of both these approaches is a **dearth of** good maps of the crustal field.

**dearth of ：**意为“缺乏、短缺”，指目前没有足够多且足够好的地壳磁场地图，导致磁导航方法难以广泛应用。它强调的是一种数量或质量上的严重不足。

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dearth 源自中古英语，本义为“饥荒、食物短缺”，后泛指任何东西的“匮乏、短缺”。

◆ 近义词组

- lack of 缺乏（最常用，中性）
- shortage of 短缺（强调供不应求）
- scarcity of 稀缺（强调稀少珍贵）
- paucity of 缺乏（非常正式，学术常用）
- insufficiency of 不足（强调不够用）
- deficiency of 缺乏、不足（常指营养、能力等）

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